

TCS-504: System Design — Assignment 1

Movie Ticket Booking System — Architectural Design & Implementation Report

1. Requirement Analysis

Functional Requirements (FR):

- **FR1 — Movie Listing:** The system shall display all movies currently marked as playing, showing title, language, and duration. If no movies are configured, an explicit empty-list message is shown without crashing.
- **FR2 — Show Listing:** Given a selected movie, the system shall list only shows scheduled for that movie, indicating screen number and start time. Movies with zero shows display an explicit 'no shows available' notification.
- **FR3 — Seat Layout Display:** Given a selected show, the system shall display every physical seat on that show's screen grouped by seat type (SILVER, GOLD, PLATINUM), annotated with AVAILABLE or BOOKED status based exclusively on that show's records.
- **FR4 — Booking Atomic Guarantee:** A customer selects one or more seat numbers. If any selected seat is already BOOKED or invalid, the entire booking is rejected with no state change. Booking is confirmed only after payment succeeds.
- **FR5 — Tiered Pricing:** Seat types map to fixed rates: SILVER ₹150, GOLD ₹250, PLATINUM ₹400. Total booking price is computed dynamically before requesting payment.
- **FR6 — Polymorphic Payment:** Supports UPI, Card, and Cash via a unified Payment interface. If payment fails, seats remain available and booking status transitions to FAILED.
- **FR7 — Ticket Generation:** Upon successful payment, a formatted ticket is printed containing Booking ID, Movie, Screen, Show Time, Seats, Amount, and CONFIRMED status.
- **FR8 — Booking Cancellation:** Given a confirmed booking ID, transitions status to CANCELLED and atomically releases all associated seats back to AVAILABLE.

Non-Functional Requirements (NFR):

- **NFR1 — Modularity:** Every class resides in its own isolated file; presentation and menu IO are decoupled from domain business logic.
- **NFR2 — Extensibility:** New payment methods (e.g. NetBanking) can be integrated by adding a single subclass of Payment without modifying BookingService or Booking (Open/Closed Principle).
- **NFR3 — Robustness:** All menu inputs and seat inputs undergo strict validation, gracefully handling invalid ranges and EOF without hanging or crashing.
- **NFR4 — Academic Scalability:** Cleanly handles at least 50 seats and 20 shows with in-memory state suitable for coursework evaluation..